Improved Upper Bounds on the Size of Condorcet Domains

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Abstract

We prove $f(n) \leq 32nf(n-1)/(11n+8)$ for the maximum size of a Condorcet domain, giving exponential base $32/11 < 3$. An exact deletion count, a decomposition into adjacent-swap components and a local counting inequality give the bound; the same local inequality gives base $8/3$ for peak-pit domains. Small examples illustrate the structural counts. The mathematical argument is presented here; the exact finite calculations, reproducible programs and complete Lean formalisation are supplied as accompanying computational material.

1 Introduction

A Condorcet domain admits rankings for which pairwise majority is transitive for every odd number of voters, allowing repeated rankings. How many rankings can be admitted while retaining consistent majority decisions? Write $f(n)$ for the maximum on n alternatives. We use Sen's equivalent combinatorial description: each triple has an alternative forbidden from one of its three positions in every admitted ranking [1]. A domain is *peak-pit* when these forbidden positions can always be chosen first or last. The alternative and the position may vary with the triple; no common axis is required. Write $h(n)$ for the peak-pit maximum.

Fishburn [2] surveys the extremal problem, and Raz [3] proved an exponential upper bound for $f(n)$. For peak-pit constructions and recent lower bounds, see Karpov and Slinko [4] and Karpov, Markström, Riis and Zhou [5]. An earlier AI-assisted working draft of this project [6] recorded the exact fibre count, the gap inequality, and bounds with general bases $56/19$ and $1336/455$, and peak-pit base $8/3$. Its arguments and verification programs were developed with Claude and examined with Codex; its finite correction coefficients were selected by linear programming. The present paper consolidates that investigation, adds the component reduction, and improves the general base to $32/11 = 2.90909\ldots$.

Theorem 1.1 (Main bounds)**.** *For every integer $n \geq 3$,*

$$f(n) \leq \frac{32n}{11n+8} f(n-1), \qquad h(n) \leq \frac{8}{3} h(n-1).$$

Consequently $f(n) \leq 4(32/11)^{n-3}$, $h(n) \leq 4(8/3)^{n-3}$, and $f(n) = O(n^{-8/11}(32/11)^n)$. The limits $\lim_{n\to\infty} f(n)^{1/n}$ and $\lim_{n\to\infty} h(n)^{1/n}$ exist and are at most $32/11$ and $8/3$, respectively.

After completion, partition the domain into components under swaps of consecutive entries. Each component is described by never-first/never-last restrictions and fixed comparisons; its deletion count treats each consecutive range of insertion positions separately. A majority projection, related to the median structures of Puppe and Slinko [7], proves this directly. Combining the component counts with the gap inequality gives the general bound. The only computational ingredient is a local inequality. The complete Lean formalisation, finite data, programs and verification instructions are provided in the accompanying repository [9].

2 Orders, triple restrictions and full families

Let X be a finite set of n alternatives. An *order* is a strict total order on X, written from first to last; $a > b$ means that a occurs before b. Restricting an order to a subset means erasing all other labels; this restriction is also called its *trace* on the subset. For a triple $T \subseteq X$ and $x \in T$, the *never-condition* $x\mathrm{N}_i$, for $i \in \{1, 2, 3\}$, says that x does not occupy position i in the restriction to T. Thus $x\mathrm{N}_1$, $x\mathrm{N}_2$ and $x\mathrm{N}_3$ mean never first, never middle and never last, respectively, *within that triple*. For instance, $2\mathrm{N}_3$ on $\{1, 2, 3\}$ permits exactly $123, 213, 231, 321$. A condition is *common* to a family if every order satisfies it. We also call a never-condition a *law*.

Definition 2.1 (Condorcet and peak-pit domains)**.** A *Condorcet domain* is a family of orders with a common never-condition on every triple. It is *peak-pit* if each triple has a common never-first or never-last condition. We call these two kinds of condition *extreme*; never-middle is the remaining kind.

By Sen's characterisation [1], this is equivalent to the voting description in the introduction. Appendix A.1 includes an elementary proof. The functions $f(n), h(n)$ are attained maxima because the set of orders is finite. A Condorcet domain is *maximal* if no further order can be added while preserving the Condorcet property; a *maximum* domain has largest possible cardinality. The first term concerns adding orders; the second concerns the largest size among all domains on the given alternatives.

Definition 2.2 (Full families and $\mathcal{F}_n$)**.** Choose one never-condition for each triple, and collect *every* order satisfying all the chosen conditions. We call the resulting family a *full domain* for that choice of conditions. To define $\mathcal{F}_n$, use the labels $0, \ldots, n-1$, choose an extreme condition on every triple, and optionally require specified pair comparisons, such as $a > b$. The nonempty families consisting of *all* orders obeying such a system form $\mathcal{F}_n$.

Here "full" refers to satisfying a chosen system exactly; it does not mean maximal. It also differs from the literature's "closed", meaning closed under majority [7]. Families on other label sets are tested for membership in $\mathcal{F}_n$ after relabelling by a bijection; all conditions and counts are invariant under this change of names.

Every maximal Condorcet domain is full: choose a common condition on each triple. All orders obeying the choices form a Condorcet enlargement, so maximality forces equality. A maximum peak-pit domain likewise equals its extreme-condition completion and belongs to $\mathcal{F}_n$. This completion need not be maximal among all Condorcet domains. Whenever we delete a label we assume $n \geq 1$.

For any family D, the *swap graph* has its orders as vertices; two are joined when exchanging consecutive entries takes one to the other. Such a swap is *enabled*. A path is a sequence of these swaps staying in D. A *swap component* consists of all orders reachable from a given order, including itself, by such paths.

3 Insertions and the exact count

For a family D on X, with $|X| = n$, define

$$D_{-x} = \{\sigma \text{ with } x \text{ erased} : \sigma \in D\}, \qquad \Sigma(D) = \sum_{x \in X} |D_{-x}|.$$

Repeated outputs are counted only once in each D_{-x}. If D is a Condorcet domain, restriction preserves its triple laws, so

$$\Sigma(D) \le nf(n-1), \qquad \Sigma(D) \le nh(n-1) \quad \text{if } D \text{ is peak-pit.} \tag{1}$$

For $P \in D_{-x}$, write $P[x : t]$ for inserting x with t labels above it; t indexes its insertion slot. Its *insertion fibre* is

$$F_D(x, P) = \{t \in \{0, \dots, n-1\} : P[x : t] \in D\}.$$

A *run* is a maximal consecutive block of occupied slots. A fibre with r runs contributes $r - 1$ to the total gap count $G(D)$; a long empty stretch between runs is one gap. Let $A(D)$ count pairs (σ, j), where $\sigma \in D$ and the swap at positions $j, j+1$, $1 \le j < n$, is enabled. Thus each edge counts twice. For a fixed family D, we abbreviate these statistics to $M = |D|,\ \Sigma = \Sigma(D),\ G = G(D),\ A = A(D)$.

The starting point is an exact count, with no inequality or lost term: every order/label choice is accounted for by a deletion output, a gap or an enabled-swap incidence.

Lemma 3.1 (Fibre identity)**.** *For every family of orders,* $\boxed{\Sigma + G + A = nM}$.

Proof. A fibre with m occupied slots and r runs has $m - r$ neighbouring occupied pairs. Summing m gives nM, and summing r gives $\Sigma + G$. Each neighbouring occupied pair is a swap edge with the inserted label distinguished. Every edge occurs twice, once for each exchanged label, giving exactly A. □

Example 3.2 (Deletion collisions)**.** The full domain $D = \{123, 213, 231, 321\}$ satisfies $2\mathrm{N}_3$. Its deletion outputs are

$$D_{-1} = \{23, 32\}, \qquad D_{-2} = \{13, 31\}, \qquad D_{-3} = \{12, 21\}.$$

For instance, deleting 1 merges $123, 213, 231$ into 23, and $F_D(1, 23) = \{0, 1, 2\}$. The swap graph is the path $123 \leftrightarrow 213 \leftrightarrow 231 \leftrightarrow 321$. Thus $M = 4, \Sigma = 6, G = 0, A = 6$, and $6 + 0 + 6 = 3 \cdot 4$.

The *top slice at x* retains only orders starting with x and then erases x. The *bottom slice at x* does the same for orders ending with x. A slice can be empty. A sum over slices runs over the choices of fixed endpoint label.

Lemma 3.3 (Two closure properties)**.** *Families in $\mathcal{F}_n$ have interval insertion fibres (one run). Every nonempty slice obtained by fixing and removing a first or last label belongs to $\mathcal{F}_{n-1}$.*

Proof. Fix the relative order of the other labels. An extreme law permits an interval of slots for the inserted label: never-first puts it below the higher of the two others, and never-last above the lower. If another label is distinguished, the law is automatic or puts the inserted label on one specified side of it. Fixed comparisons also give intervals. Intersect these intervals and use fullness. For an endpoint slice, internal triple laws remain. A triple containing the removed endpoint reduces to a fixed comparison, a tautology or a contradiction. Pair constraints reduce similarly; contradictions mean the slice is empty. The remaining constraints describe the slice exactly. □

4 Reducing a general domain to extreme laws

Throughout this section D is full. Its swap components are the connected pieces of the graph defined in Section 2. For three orders write $\mathrm{maj}(\alpha, \beta, \gamma)$ for their pairwise majority order. The majority-closure argument is part of the median structure studied by Puppe and Slinko [7]; we give the specific adjacent-swap component argument in full.

Lemma 4.1 (Majority projection)**.** *The majority of three members of D lies in D. If K is a swap component of D and U is the set of all oriented pair comparisons common to every order of K, then*

$$K = \{\tau \in D : \tau \text{ satisfies every comparison in } U\}.$$

Proof. Majority is an order by Appendix A.1. On a triple $\{x, y, z\}$, the law $x\mathrm{N}_2$ says $x > y$ if and only if $x > z$; the two majority comparisons therefore agree as well. If majority violated a never-first law, two sets of at least two voters among three would support its two comparisons. Their intersection would contain a voter violating the law. Never-last is symmetric. Fullness therefore puts the majority order in D.

For the displayed equality, suppose $\tau \in D$ satisfies U. Choose $\sigma \in K$ disagreeing with τ on as few pair comparisons as possible. If they differ, a differing comparison is not fixed on K. Take $\rho \in K$ agreeing with τ there, and project a path from σ to ρ by $v \mapsto \mathrm{maj}(\sigma, \tau, v)$. Images lie in D; consecutive images differ in at most one comparison. Two distinct orders differing in one comparison exchange adjacent labels: an intervening label would force another comparison to change. Remove consecutive repeated images; the remaining path stays in K and starts at σ. On comparisons where σ, τ agree it is constant; elsewhere it follows v. Its last order corrects the selected disagreement without losing any agreement. This contradicts the choice of σ, proving the claim. □

Proposition 4.2 (Component identity)**.** *Every component K belongs to $\mathcal{F}_n$, and*

$$\sum_K |K| = M, \qquad \sum_K \Sigma(K) = \Sigma(D) + G(D). \tag{2}$$

Proof. A chosen never-middle law $x\mathrm{N}_2$ equates the two comparisons of x on that triple. Neither can change along a swap path, which changes only one comparison at a time. Thus throughout K, x is above both others or below both. Replace the law by $x\mathrm{N}_3$ or $x\mathrm{N}_1$, respectively. Keep the original extreme laws and add all comparisons U fixed on K. Every order in K satisfies these constraints. Conversely, they recover the old never-middle laws, so an order satisfying them lies in D and agrees with U. Lemma 4.1 puts it in K. Thus these conditions describe K exactly, proving $K \in \mathcal{F}_n$.

Components partition D, giving the first identity. Each original fibre run lies in one component, because consecutive occupied slots are swap edges. A component cannot contain two separated runs: its own fibre is an interval by Lemma 3.3. Thus each original run gives exactly one component deletion output. Summing runs gives the second identity. □

Example 4.3 (Counting separated runs)**.** The full domain $D = \{123, 132, 231, 321\}$, defined by $1\mathrm{N}_2$, has components $\{123, 132\}$ and $\{231, 321\}$. Its 1-fibres above 23 and 32 both have slots $\{0, 2\}$, each contributing one gap. Thus $\Sigma(D) = 6, G(D) = 2$. Each component has deletion sum 4, so $4 + 4 = 6 + 2$: splitting the family counts its separated insertion runs separately.

5 The gap inequality

Fix a Condorcet domain D on X, with $|X| = n$. A *lift* moves one label upwards, and a *drop* moves it downwards, keeping all other relative positions and remaining in D. These moves may jump several slots. An incidence (σ, x) is *free* if it has both moves, and *frozen* if it has neither; their counts are $\mathrm{fr}(D), \mathrm{fz}(D)$. In an order σ, consecutive entries a, c form a *locked* pair if a has no drop and c has no lift. Let $\mathrm{lk}(D)$ count these triples (σ, a, c). With D fixed, abbreviate these counts to $\mathrm{fr}, \mathrm{fz}, \mathrm{lk}$.

Lemma 5.1 (Two counts)**.** *For a full domain,*

$$2\Sigma = nM - \mathrm{fr} + \mathrm{fz} = (n+1)M - A + \mathrm{lk}, \qquad 3\Sigma + \mathrm{fr} = (n+1)M + G + \mathrm{fz} + \mathrm{lk}\,. \tag{3}$$

Proof. Each fibre has one order with no lift and one with no drop. Their combined count is 2Σ; an incidence contributes 0, 1 or 2 according as it is free, has one move, or is frozen. For an adjacent pair, the missing drop of its upper label and missing lift of its lower label contribute 0 if the swap is enabled, 2 if locked, and 1 otherwise. Indeed a blocked swap changes only that pair comparison and violates a chosen law on a triple containing the pair. Write its old trace as zac if the third label is above. The violated law is $a\mathrm{N}_3$ or $c\mathrm{N}_2$, either of which forbids every drop of a. If the third label is below, the old trace is acz; the law is $a\mathrm{N}_2$ or $c\mathrm{N}_1$, forbidding every lift of c. Thus both crossing moves cannot be available. The first label also has no lift and the last no drop. The second count is therefore $2M + (n-1)M - A + \mathrm{lk}$. Combine the two counts with $\Sigma + G + A = nM$ for the final identity. □

Lemma 5.2 (Free-to-locked injection)**.** *For a maximal Condorcet domain,* $\mathrm{fr} \le \mathrm{lk}$.

Proof. First, if $a > c$ in every order, they cannot be adjacent in one. Swapping such an occurrence adds an order while preserving a common law on every triple: triples containing the pair retain $c\mathrm{N}_1$ if the third label is above, or $a\mathrm{N}_3$ if below. All other triples are unchanged. This contradicts maximality.

A free label b is neither first nor last. Write a, b, c for the consecutive block centred on b. The order, a lift and a drop give abc, bac, acb; their only possible common laws are $a\mathrm{N}_3$ and $c\mathrm{N}_1$. Both cannot hold, since together they force $a > c$ everywhere, whereas a lift puts them consecutively. If $a\mathrm{N}_3$ holds, take the available lift leaving b as low as possible; otherwise take the available drop leaving it as high as possible: use the closest available slot in the chosen direction. In the image order the pair a, c is consecutive and locked: either crossing move would violate the selected law. Map the free incidence to this order/pair incidence. For a specified moved label, the preimage is recovered uniquely by putting it back between the image pair. Appendix A.2 excludes collisions between different moved labels, completing the proof. □

The first identity in Lemma 5.1 is the sum over deleted labels of Fathi's fibre-excess identity [8, eq. (96)]. The two possible laws at a free incidence are also Fathi's flank-rule observation [8, Lemma K.52]. The injection uses maximality to make the assignment collision-free.

Corollary 5.3 (Gap inequality)**.** *For every maximal Condorcet domain,* $G + (n+1)M \le 3\Sigma$.

Proof. In the last identity of (3), use $\mathrm{fr} \le \mathrm{lk}$ and $\mathrm{fz} \ge 0$. □

6 A precise local inequality

The next ingredient concerns seven consecutive positions. Its correction terms depend only on six-label endpoint slices and cancel when windows overlap. Exact data and exhaustive checks are supplied in the computational material [9]. Recall that top and bottom slices fix and delete the first and last label, respectively. Empty slices contribute zero. For $J \in \mathcal{F}_7$, let $a_j(J)$ count orders whose swap at positions $j, j+1$ is enabled, for $1 \le j \le 6$. Define

$$C(J) = \frac{a_1(J) + 2a_2(J) + 2a_3(J) + 2a_4(J) + 2a_5(J) + a_6(J)}{10}. \tag{4}$$

Summing $A(H)$ over top and bottom six-slices counts the end swaps once and all other swaps twice; hence that sum is $10C(J)$.

Lemma 6.1 (Seven-position inequality)**.** *There is a fixed rational-valued function q, defined below, such that every $J \in \mathcal{F}_7$, and the empty family, satisfies*

$$C(J) \le \frac{5}{8}|J| + \sum_{\text{bottom } H} q(H) - \sum_{\text{top } H} q(H). \tag{5}$$

On five labels, enumerate the families in $\mathcal{F}_5$, identify families related by a bijective renaming of labels (*isomorphic* families), and call each such class a *type*. Encode a family by a binary *mask*: list all permutations in lexicographic order, starting at index 0, and set bit i exactly when permutation i belongs to the family. Order the types by their least masks over all label renamings. There are 1160 types, indexed $0, \dots, 1159$. Set $\phi(P) = W_k/d$ on type k, where $d = 8\,846\,117\,280$. The complete coefficient table and its exhaustive generation are supplied with the computation [9].

For a six-label family $H \in \mathcal{F}_6$, define

$$b(H) = \sum_{\text{bottom } P} |P|\phi(P), \qquad t(H) = \sum_{\text{top } P} |P|\phi(P),$$
$$s(H) = \frac{12}{19}|H| + b(H) - t(H) - \frac{1}{5}A(H), \qquad \eta = \frac{1}{152}.$$

Let $r_b(H)$ count bottom five-slices of type 866, and $r_t(H)$ top five-slices of type 41. The accompanying table specifies fourteen exceptional six-types: five assigned to $\mathcal{T}$ and nine to $\mathcal{S}$. Both classes include every relabelled copy. The nine integer corrections define $dp(H)$ on $\mathcal{S}$. Outside $\mathcal{S}$, put $p(H) = 2\eta(r_t(H) - r_b(H))$. Finally set

$$\theta_b(H) = s(H)/2 + p(H), \qquad \theta_t(H) = s(H)/2 - p(H), \qquad q(H) = (b(H) + t(H))/2 + p(H).$$

All these quantities are zero on the empty family. Arguments are suppressed when a single slice H is under discussion.

The following facts are verified by exhaustive exact computation; the coverage proof and code are supplied separately [9].

F1. There is a unique eight-order family $R_8 \in \mathcal{T}$ up to relabelling. All its copies H have $\theta_b(H) = \theta_t(H) = 0$.

F2. Outside $\mathcal{T} \cup \mathcal{S}$, $\theta_b \ge \eta|H|/2 + 4\eta r_t$ and $\theta_t \ge \eta|H|/2 + 4\eta r_b$.

F3. If J has a six-slice in $\mathcal{S} \cup \mathcal{T}$ that is not isomorphic to R_8, then

$$\Lambda(J) := \sum_{\text{bottom } H} \theta_b(H) + \sum_{\text{top } H} \theta_t(H) \ge \eta|J|.$$

7 Paying for the remaining slices

We now prove the local inequality from the explicitly checked finite facts; the remaining argument is by hand. A convenient labelled representative of the eight-order exceptional type is

$$R_8 = \{524130, 524310, 534120, 534210, 542130, 542310, 543120, 543210\}.$$

Its common first and last labels are 5 and 0. Deleting the first gives five-type 866, and deleting the last gives type 41. Its only nonempty bottom and top slices therefore have types 41 and 866, respectively, so $r_b(R_8) = r_t(R_8) = 0$. These statements and its equality with the supplied eight-order type are also verified by the indexing procedure supplied with the computation.

Proof of Lemma 6.1. F3 already gives $\Lambda(J) \geq \eta|J|$ unless all nonempty six-slices are regular (outside $\mathcal{T} \cup \mathcal{S}$) or of type R_8. Consider this remaining case.

Each regular slice contributes a baseline $\eta|H|/2$ by F2. An R_8 slice contributes zero by F1 and is missing baseline 4η. Suppose a bottom R_8 slice is obtained by fixing last label z. Its orders share first label w, and therefore also enter the top slice at w. This top receiver, upon fixing and deleting its last label z, has a five-slice of type 866. Thus its r_b is positive. It cannot be R_8, whose r_b is zero, so it is regular. F2 supplies an extra 4η in its θ_t, exactly the missing baseline.

Different bottom R_8 slices have different last labels. They charge different receiver/last-label pairs, each counted separately in r_b, so no contribution is reused. Reverse top and bottom for top R_8 slices; their receivers use the separate θ_b contributions from r_t. All missing baselines are paid. At each end the slice sizes sum to $|J|$, yielding total baseline $\eta|J|$. Thus the bound on Λ holds in every case.

It remains to expand the definitions. Since the slice sizes sum to $|J|$ at each end,

$$\begin{aligned}\Lambda(J) = \frac{12}{19}|J| - \frac{1}{10} \sum_{\text{top and bottom } H} A(H) \\ + \frac{1}{2} \sum_{\text{bottom and top } H} (b(H) - t(H)) + \sum_{\text{bottom}} p(H) - \sum_{\text{top}} p(H).\end{aligned}$$

The sum of $t(H)$ over bottom slices equals the sum of $b(H)$ over top slices. Both enumerate the same conditional five-families obtained by fixing one first and one last label and deleting them. The deletions commute, including their label normalisations. Hence

$$\Lambda(J) = \frac{12}{19}|J| - C(J) + \sum_{\text{bottom}} q(H) - \sum_{\text{top}} q(H).$$

Use $\Lambda(J) \geq \eta|J|$ and $12/19 - 1/152 = 5/8$. For the empty family every term is zero. □

The exceptional case is therefore resolved by a counting argument, rather than by checking all seven-label families. The computation handles only the thirteen other exceptional six-types and their two endpoint orientations. The accompanying computation gives a full coverage proof.

8 From local windows to an exact density

Proposition 8.1. *For every $n \geq 1$ and $K \in \mathcal{F}_n$, $8A(K) \leq 5n|K|$ and $\Sigma(K) \geq 3n|K|/8$.*

Proof. First suppose $n \geq 7$, and write $M = |K|$. At each block of seven consecutive positions, fix the complete ordered prefix and suffix outside it. Its conditional family J belongs to $\mathcal{F}_7$ when nonempty, by repeated endpoint slicing. At each window position these families partition K, so their sizes sum to M. A swap within a window has the same availability in its conditional family as in K. Apply (5) to all contexts and windows.

The top six-slices of one window are exactly the bottom six-slices of the next: each fixes the same ordered exterior and retains the same central six labels. For example the windows $xabcdef$ and $abcdefy$ share $abcdef$, with x and y fixed outside after slicing. Thus all internal q-terms cancel. This matching retains the full exterior, not merely the isomorphism type of a slice.

The class $\mathcal{F}_6$ is finite, so $Q = \max(\{0\} \cup \{|q(H)|/|H| : H \in \mathcal{F}_6\})$ is finite. Thus $|q(H)| \leq Q|H|$, also for the empty family. Each of the two surviving outside layers has total size M, so its absolute contribution is at most QM. Consequently

$$\sum_{\text{all window contexts } J} C(J) \leq \frac{5}{8}(n-6)M + 2QM.$$

A swap contained in six windows has total weight $(1+2+2+2+2+1)/10 = 1$. Every swap has weight between zero and one. The sum of all position weights is $n-6$, whereas there are $n-1$ swap positions. The total deficit is exactly five per order. Since each swap indicator is at most one,

$$A(K) \leq \frac{5}{8}nM + BM, \qquad B = 5/4 + 2Q. \tag{6}$$

Fix any $K \in \mathcal{F}_n$, now allowing small n, and concatenate t disjoint copies in a fixed block order. Each block chooses an order independently; all comparisons between blocks are fixed. A mixed triple has a label fixed first or last, so compatible extreme laws can be chosen. The resulting full family $K^{[t]}$ belongs to $\mathcal{F}_{tn}$, and

$$|K^{[t]}| = M^t, \qquad A(K^{[t]}) = tA(K)M^{t-1}.$$

The second identity counts an enabled swap in any one block and independent choices in the others. A swap across a join reverses a fixed comparison and is unavailable; it creates no new term. For $tn \geq 7$, apply (6) to the joined family:

$$tA(K)M^{t-1} \leq \frac{5}{8}tnM^t + BM^t, \qquad A(K) \leq \left(\frac{5}{8}n + \frac{B}{t}\right)M.$$

Let t grow. This proves $A(K) \leq 5nM/8$, including the original small n. Its fibres are intervals, so $G = 0$; Lemma 3.1 then gives the deletion bound. □

Example 8.2 (Two joined copies)**.** Join the four-order domain D of Example 3.2 to its relabelled copy $D' = \{456, 546, 564, 654\}$, always putting its first block above its second, and call the joined family $J \in \mathcal{F}_6$. It has 16 orders. Each deletion leaves two choices in its affected block and four in the other, so $\Sigma(J) = 6 \cdot 2 \cdot 4 = 48$. The internal paths give $A(J) = 2 \cdot 6 \cdot 4 = 48$, and $G(J) = 0$. The join adds no swap: crossing it reverses a fixed comparison.

9 The upper bounds and their scope

Proof of Theorem 1.1. Let D be a maximum Condorcet domain on n labels, with $M = f(n)$. It is maximal and therefore full. By the component identity and Proposition 8.1,

$$\frac{3}{8}nM \le \sum_K \Sigma(K) = \Sigma + G.$$

The gap inequality gives $\Sigma + G \le 4\Sigma - (n+1)M$. Combining,

$$\frac{3}{8}nM \le 4\Sigma - (n+1)M, \qquad (11n+8)M \le 32\Sigma \le 32nf(n-1).$$

This proves the general recurrence. A maximum peak-pit domain belongs to $\mathcal{F}_n$, so its deletion sum is at least $3nM/8$ and at most $nh(n-1)$, proving the peak-pit recurrence.

On three labels a common law omits two of the six orders; the four-order domain in Example 3.2 attains four. Thus $f(3) = h(3) = 4$. Iterating the recurrences gives both geometric bounds, since $32n/(11n+8) \le 32/11$. More precisely, with $r = 32/11$ and $a = 8/11$,

$$f(n) \le 4r^{n-3} \prod_{k=4}^{n} \frac{k}{k+a} \le 4r^{n-3} \left(\frac{4}{n+1}\right)^a.$$

For the last step, $(1+u)^a \le 1+au$ for $u \ge 0$: $1+au-(1+u)^a$ is zero at zero and has nonnegative derivative. Thus $k/(k+a) \le (k/(k+1))^a$, and the product telescopes. This proves the claimed polynomial correction.

Finally, concatenate two domains in a fixed block order. Majority is transitive within each block and respects the block order; mixed triples have a fixed extreme label. Thus both the Condorcet and peak-pit properties are preserved, and cardinalities multiply. Hence for $g = f$ or h, $g(m+n) \ge g(m)g(n)$. Singletons give $g(n) \ge 1$. Put $a_n = \log g(n) \ge 0$; the unique empty order gives $g(0) = 1$. For fixed $k \ge 1$, write $n = qk + s$, $0 \le s < k$. Then $a_n \ge qa_k + a_s \ge qa_k$, so $\liminf a_n/n \ge a_k/k$. Taking the supremum over k shows that the liminf is at least $\sup_k a_k/k$; the limsup cannot exceed it. The geometric bound makes this supremum finite and bounds the limit by $\log r$, or $\log(8/3)$. Exponentiating proves the root-growth assertions. □

Computation and formal verification

The structural counts, component decomposition, receiver argument and limiting steps are hand proofs; Appendix A.2 supplies all collision cases of the injection. The sole computational input is the finite comparison in Section 6. Its coefficients, exhaustive coverage argument, exact integer checks and reproduction instructions are supplied with the paper and in the repository [9]. All mathematical comparisons use exact integers and no optimiser.

All seven conclusions of Theorem 1.1 are also proved end to end in Lean 4.26.0 with Mathlib 4.26.0. The entry point is `CDCert.Main`; the local inequality is `CDCert.K6.finite_lemma`. Finite certificates and their coverage are checked by Lean's kernel: a successful Python run is not a premise of the formal proof. The final results use no unproved placeholders or additional axioms, only Lean's standard `propext`, `Classical.choice` and `Quot.sound`. The repository records the exact build, theorem mapping and verification scope [9]; this verifies the mathematical statements, not historical claims or the Python programs themselves.

Claude and ChatGPT/Codex assisted this work; the mathematical claims and their presentation remain the authors' responsibility.

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A Voting interpretation and the injection

A.1 Equivalence with the voting definition

A voter profile is a finite list of admitted orders, with repetitions allowed. Pairwise majority puts a above b if more than half the voters do. We prove that Definition 2.1 is equivalent to transitivity of majority for every odd positive profile size.

On a triple, each of $\{abc, bca, cab\}$ and $\{acb, cba, bac\}$ gives a majority cycle. For a voting Condorcet domain, at least one order in each set must therefore be missing from its triple restrictions. By cyclic relabelling let the first missing order be abc. Pairing it with acb, cba, bac gives, respectively, $a\mathrm{N}_1, b\mathrm{N}_2, c\mathrm{N}_3$ on all the remaining orders. Conversely, under $x\mathrm{N}_2$ the two comparisons of x agree for each voter, so they agree for majority, excluding a cycle. Under $x\mathrm{N}_1$, each voter with $x > y$ has $z > x > y$. A majority for $x > y$ therefore forces a majority for $z > y$, excluding $x > y > z > x$. Exchange y, z for the reverse cycle, and reverse orders to treat $x\mathrm{N}_3$. An odd majority compares every distinct pair in exactly one direction. If it is not transitive, there are $a > b$, $b > c$, but $c > a$: a directed triangle. Thus excluding cycles on triples proves transitivity. This proves the equivalence attributed to Sen [1].

A.2 The injection: all collision cases

We complete the hand proof of Lemma 5.2. Let the target locked pair be p, q in its image order τ. For a preimage with moved label b, call it *upper* if b lies above p in τ, and *lower* otherwise. For an upper preimage, D has common law $p\mathrm{N}_3$ on $\{p, q, b\}$, whereas $q\mathrm{N}_1$ is not common to D. There is consequently an order χ_b with trace qpb: the alternative qbp would violate $p\mathrm{N}_3$. For a lower preimage, $q\mathrm{N}_1$ is common to D, and $p\mathrm{N}_3$ is not. The original neighbours in either preimage are p, b, q.

Two upper preimages

Suppose different moved labels b_1, b_2 give the same target. Let their preimage orders be σ_1, σ_2, and drop b_i in σ_i to obtain ρ_i; such drops exist because the preimages are free. Their traces on $\{b_1, b_2, q\}$ are

order	σ_1	σ_2	ρ_1	ρ_2
trace	b_2b_1q	b_1b_2q	b_2qb_1	b_1qb_2.

Both b_1, b_2 occupy all three positions among these orders; q occupies the last two. Thus the only common law is $q\mathrm{N}_1$. In χ_{b_1}, q is above p, b_1. The law just found forces b_2 above q, giving b_2qp on $\{p, q, b_2\}$, contrary to its law $p\mathrm{N}_3$.

Two lower preimages

Reverse every order. Majority transitivity and maximality are preserved. Lifts and drops exchange roles, and the unique law used by the assignment changes from never-first to never-last. The highest drop becomes the lowest lift. Thus the reversed assignment has two upper preimages, which the preceding argument excludes.

One upper and one lower preimage

Let b_1 be upper and b_2 lower. On the four indicated labels,

$$\tau : b_1pqb_2, \qquad \sigma_1 : pb_1qb_2, \qquad \sigma_2 : b_1pb_2q.$$

Take $\chi = \chi_{b_1}$, whose trace on $\{p, q, b_1\}$ is qpb_1. The lower preimage's law $q\mathrm{N}_1$ on $\{p, q, b_2\}$ forces b_2 above q, so χ has trace b_2qpb_1. On $\{b_1, b_2, q\}$, the orders τ, σ_2, χ give

$$b_1qb_2, \qquad b_1b_2q, \qquad b_2qb_1.$$

These allow exactly two common laws: $b_1\mathrm{N}_2$ and $q\mathrm{N}_1$. Drop b_1 from σ_1 to obtain ρ. Now q lies above b_1, and its comparison with b_2 is unchanged, so it lies above both. This fourth witness excludes $q\mathrm{N}_1$, leaving $b_1\mathrm{N}_2$. It forces the trace of ρ to be qb_2b_1.

Lift b_2 from σ_2 to obtain λ. Its comparison with q is now upwards and the comparison of b_1, q is unchanged. The same never-middle law forces its trace on $\{b_1, b_2, q\}$ to be b_1b_2q. On $\{p, b_1, b_2\}$, the five orders $\tau, \sigma_1, \rho, \lambda, \chi$ therefore give

$$b_1pb_2, \quad pb_1b_2, \quad pb_2b_1, \quad b_1b_2p, \quad b_2pb_1.$$

Each label occupies every position, leaving no common never-law. This contradicts Definition 2.1. All collisions are excluded, proving injectivity. □